

HashMap ==> store the data in the form of key value pairs

Construct the HashMap to store the student name and cgpa as(key, value) is for n students and print all the details using keySet()

keys cannot be duplicated, values can be duplicated
{ b: 10, a: 15, c: 10 }
get(c)

[10, 10, 20, 30, 10, 40, 50, 20]

count the occurences of each element resnt in the array list

10 ==> 3
20 ==> 2
30 ==> 1
40 ==> 1
50 ==> 1

[10, 10, 20, 30, 10, 40, 50, 20]

A square box with a black border containing handwritten text in brown ink. The text is arranged in two lines: "10:2" on the top line and "20:1" on the bottom line.

Given n, find the last digit of the nth fibonacci using pisano period

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 243,
1 2 3 4 5 6 7 8 9 10 ~~~~~

n = 1,00,000

pisano period ==> hel to identify any fibonaci no at any nth value in efficient way

Monday, Tuesday, Sunday
1st Aug

Monday
8th Aug

Monday
15th Aug

Monday
22nd Aug

0, 1, 1, 2, 3, 5, 8, **13**, **21**, **34**, **55**, **89**, **144**, **243**,
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

1st to 60th ==> 0,1,1,2,3,5.....
61st to 120 ==> 0,1,1,2,3,5.....

1000 th position ==> fibonnaci ==> last digit

0th ==> 0
60th ==> 0
120th ==> 0
.....
600th ==> 0
720th ==> 0
840th ==> 0

960th ==> 0
961th ==> 1
962th ==> 1
963 ==> 2
964 ==> 3

1000 / 60 ==>
1000 % 60